

Macroeconomics II Part II, Lecture V: The New-Keynesian Model: Shocks & Propagation

Erik Öberg

Recap

- Basic NK model = RBC model + monopolistic competition and sticky prices
- Last lecture: Setup, derivations and determinacy
- Today: What are the predictions of the NK model?

Agenda

- 1 Monetary policy shocks
- 2 TFP shocks
- 3 Cost-push shocks

Monetary policy shocks

The 3-equation representation

- The log-linearized equilibrium can be characterized by

$$\begin{aligned}\text{DIS curve:} \quad & \hat{y}_t = -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{y}_{t+1} \\ \text{Phillips curve:} \quad & \pi_t = \beta E_t \pi_{t+1} + \kappa \hat{y}_t \\ \text{Policy rule:} \quad & \hat{i}_t = \phi \pi_t + \nu_t\end{aligned}$$

- 3 equations in 3 unknowns: $\{\hat{y}_t, \hat{i}_t, \pi_t\}$!
- This is how the model is usually presented in the literature
- **Warning!**: Although very convenient, these equations mix multiple equilibrium relationships $\Rightarrow$ hard to extract a precise intuition about model mechanisms

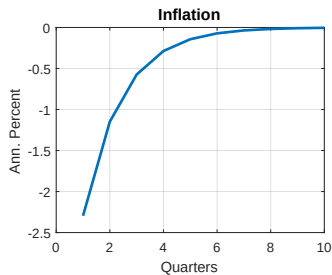
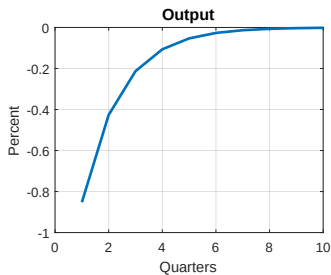
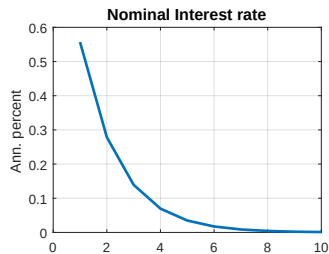
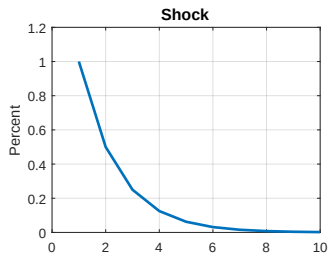
Calibration

- Quarterly frequency
- For the recurrent parameters, let's stick with what we used for our RBC model
 - ▶ $\beta = 0.99$
 - ▶ $\varphi = 1$
- The new ones:
 - ▶ $\theta = 2/3$ to match an average price duration of three quarters (Galí-Lopéz-Salido EER 2001)
 - ▶ $\epsilon = 6$ to match average markup of 20%
 - ▶ $\phi = 1.5$ to match the Fed reaction function during the Greenspan era
- The shock process:

$$\nu_t = \rho_\nu \nu_{t-1} + \epsilon_t$$

with $\rho_\nu = 0.5$ to generate a “moderately” persistent shock, as we saw in the empirical IRFs

IRFs to a monetary policy shock



Comments I

- In contrast to the vanilla RBC model, there are no humps and bumps here
- This is because the model has no state variable, so dynamics are completely forward-looking:

$$\begin{aligned}\text{DIS curve:} \quad & \hat{y}_t = -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{y}_{t+1} \\ \text{Phillips curve:} \quad & \pi_t = \beta E_t \pi_{t+1} + \kappa \hat{y}_t \\ \text{Policy rule:} \quad & \hat{i}_t = \phi \pi_t + \nu_t\end{aligned}$$

- This means that the policy function for any variable $x_t \in \{\hat{y}_t, \pi_t, \hat{i}_t\}$

$$x_t = \sum_{s=0}^t a_{t-s}^x \nu_s$$

has $a_{t-s} = 0$ for all $s < t$, i.e.,

$$x_t = a_0^x \nu_t.$$

- An implication is that you can solve for the IRFs analytically, see problem set 6.

- Basic NK model qualitatively matches the evidence: a surprise increase in the policy rate leads to a fall in y
- Quantitatively, it is far off; we will get back to this in Lecture VI
- To understand the mechanism, let's go back to the full 8-equation system, where we have cleaner interpretations of the equilibrium equations

Reminder: Full system looks like...

- The log-linearized equilibrium is characterized by

$$\text{Intratemporal hh optimality: } \hat{\omega}_t = \hat{c}_t + \varphi \hat{n}_t$$

$$\text{Intertemporal hh optimality: } \hat{c}_t = -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{c}_{t+1}$$

$$\text{Firm optimality: } \pi_t = \beta E_t \pi_{t+1} + \lambda \widehat{mc}_t$$

$$\text{Marginal cost: } \widehat{mc}_t = \hat{\omega}_t$$

$$\text{Goods clearing: } \hat{c}_t = \hat{y}_t$$

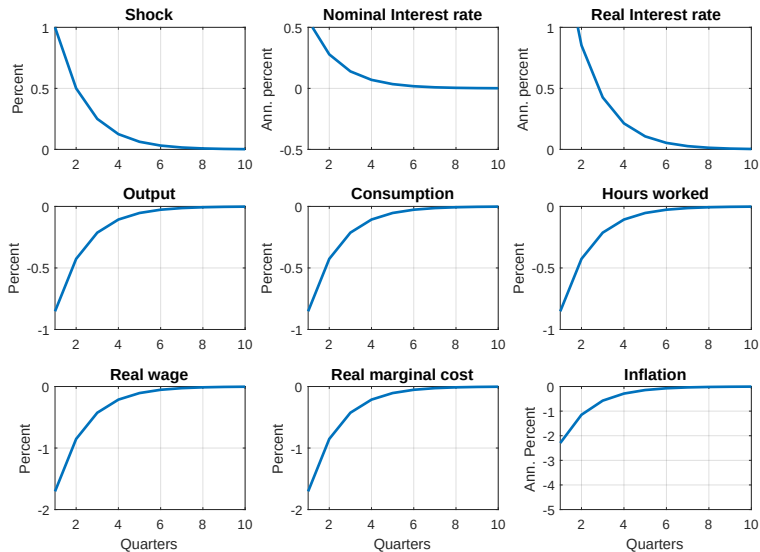
$$\text{Labor clearing: } \hat{y}_t = \hat{n}_t$$

$$\text{Policy: } \hat{i}_t = \phi \pi_t + \nu_t$$

where $\hat{\omega}_t = \hat{w}_t - p_t$ denotes log deviations in the real wage

- Note: Real interest rate given by $\hat{r}_t = \hat{i}_t - E_t \pi_{t+1}$

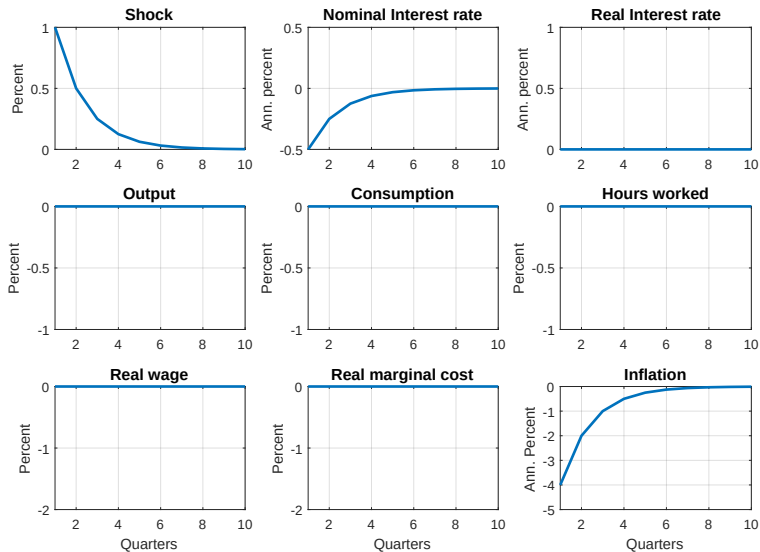
IRFs to a monetary policy shock: full system



IRFs to a monetary policy shock: mechanism

- How can we explain the equilibrium responses?
- Simplify, simplify, simplify
- Let's start with flexible prices: $\theta \rightarrow 0$

IRFs to a monetary policy shock under flexible prices



Mechanism under flexible prices

- Last lecture, we showed that under flexible prices, system collapses to

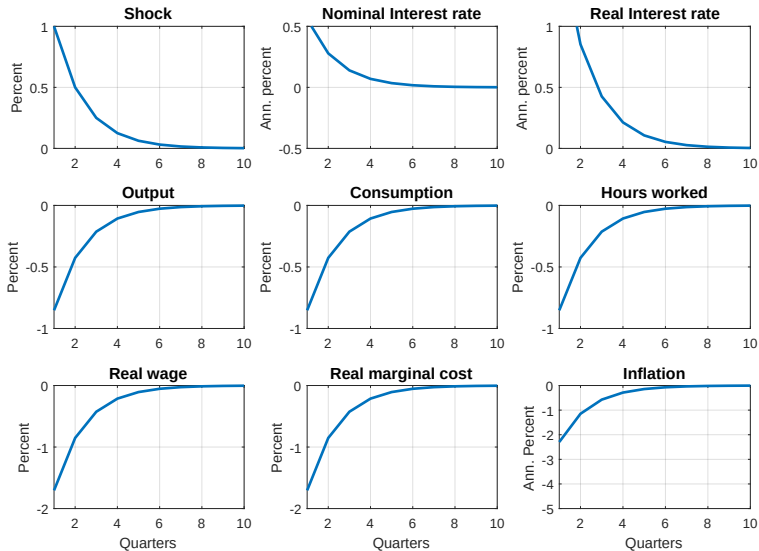
$$\begin{aligned}\text{DIS curve:} \quad & \hat{i}_t = E_t \pi_{t+1} \\ \text{Policy rule:} \quad & \hat{i}_t = \phi \pi_t + \nu_t\end{aligned}$$

- Zero response of all real variables, i.e., **monetary neutrality**
- We also showed that the unique solution has

$$\pi_t = - \sum_{s=0}^{\infty} \frac{1}{\phi^{s+1}} \nu_{t+s}$$

- Inflation falling in response to a “contractionary” monetary policy shock has nothing to do with sticky prices or other frictions; it is simply a consequence of a Taylor rule and rational expectations
- The inflation response is so strong that the interest rate falls in response to a positive shock!

Back to sticky prices



- Sticky prices $\rightarrow$ smaller inflation response
- With smaller inflation response, nominal interest rate actually increases
- As a consequence, the real interest rate increases:

$$\hat{r}_t = \hat{i}_t - E_t \pi_{t+1}$$

- Monetary non-neutrality!
- Given that the real interest rate increases, we can back out the response of all other variables
- In so doing, recall that we have assumed that we only search for bounded solutions!

- Take as given that $\hat{r}_t$ increases, then

$$\text{Intertemporal hh optimality: } \hat{c}_t = -(\hat{r}_t) + E_t \hat{c}_{t+1}$$

implies $\Delta E_t c_{t+1}$ is positive

- $\Delta E_t c_{t+1} > 0$ + Bounded solution $\Rightarrow \{\hat{c}_{t+s}\}$ must converge to some steady state
- But the steady state is unique, so we know $\{\hat{c}_{t+s}\}$ must converge to $\hat{c}_{t+s} = 0$
- Therefore, we must have that $\hat{c}_t < 0$!

- Take as given that $\hat{c}_t < 0$
- Market clearing $\hat{c}_t = \hat{y}_t = \hat{n}_t < 0$
 - ▶ $\Rightarrow$ we may think of the output drop as being caused by a drop in **aggregate demand**
- How is this consistent with optimal labor supply? Intratemporal optimality condition:

$$\text{Intratemporal hh optimality:} \quad \hat{\omega}_t = \hat{c}_t + \varphi \hat{n}_t$$

- $\hat{n}_t < 0$ only if $\hat{\omega}_t < \hat{y}_t < 0$
 - ▶ Wages need to respond more than output (and profits less)!
- Is the fall in inflation consistent with the Phillips curve?
 - ▶ $\hat{\omega}_t < 0 \Rightarrow \hat{m}c_t < 0 \Rightarrow \beta E_t \pi_{t+1} - \pi_t \approx \Delta \pi_{t+1} > 0$ from the Phillips curve
 - ▶ Again, bounded solution + unique steady state $\Rightarrow \hat{\pi}_t < 0$

A monetary policy shock is equivalent to a demand shock

- Recall: DIS curve stems from household Euler equation

$$c_t = -(i_t - E_t \pi_{t+1} - \xi) + E_t c_{t+1}$$

where $\xi = -\log \beta$

- Suppose there are shocks to discount factor β
 - Specifically, assume $\xi_t = \rho_\xi \xi_{t-1} + \epsilon_{\xi,t}$
 - = shock to the marginal value of current consumption - a “demand shock”
- Then, 3-equation system becomes

$$\text{DIS curve: } \hat{y}_t = -(\hat{i}_t - E_t \pi_{t+1} - \xi_t) + E_t \hat{y}_{t+1}$$

$$\text{Phillips curve: } \pi_t = \beta E_t \pi_{t+1} + \kappa \hat{y}_t$$

$$\text{Policy rule: } \hat{i}_t = \phi \pi_t + \nu_t$$

or

$$\text{DIS curve + Policy rule: } \hat{y}_t = -(\phi \pi_t - E_t \pi_{t+1}) + E_t \hat{y}_{t+1} + \xi_t - \nu_t$$

$$\text{Phillips curve: } \pi_t = \beta E_t \pi_{t+1} + \kappa \hat{y}_t$$

- $\Rightarrow$ Positive monetary policy shocks are equivalent to negative demand shocks

TFP shocks

TFP shocks

- So far: focus on monetary policy shocks
- Naturally, we are interested in how sticky prices and the behavior of monetary policy affect the response to non-policy shocks
- Let's consider TFP shocks
- Assume that $a_t \equiv \log A_t$ follows

$$a_t = \rho_a a_{t-1} + \epsilon_t^a$$

- Which equations in our equilibrium system are affected?

Intratemporal hh optimality: $\hat{\omega}_t = \hat{c}_t + \varphi \hat{n}_t$

Intertemporal hh optimality: $\hat{c}_t = -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{c}_{t+1}$

Firm optimality: $\pi_t = \beta E_t \pi_{t+1} + \lambda \widehat{mc}_t$

Marginal cost: $\widehat{mc}_t = \hat{\omega}_t$

Goods clearing: $\hat{c}_t = \hat{y}_t$

Labor clearing: $\hat{y}_t = \hat{n}_t$

Policy: $\hat{i}_t = \phi \pi_t + \nu_t$

TFP shocks

- So far: focus on monetary policy shocks
- Naturally, we are interested in how sticky prices and the behavior of monetary policy affect the response to non-policy shocks
- Let's consider TFP shocks
- Assume that $a_t \equiv \log A_t$ follows

$$a_t = \rho_a a_{t-1} + \epsilon_t^a$$

- Which equations in our equilibrium system are affected?

Intratemporal hh optimality: $\hat{\omega}_t = \hat{c}_t + \varphi \hat{n}_t$

Intertemporal hh optimality: $\hat{c}_t = -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{c}_{t+1}$

Firm optimality: $\pi_t = \beta E_t \pi_{t+1} + \lambda \widehat{mc}_t$

Marginal cost: $\widehat{mc}_t = \hat{\omega}_t$

Goods clearing: $\hat{c}_t = \hat{y}_t$

Labor clearing: $\hat{y}_t = \hat{n}_t$

Policy: $\hat{i}_t = \phi \pi_t + \nu_t$

Incorporating TFP shocks

- Marginal cost:

$$MC_t = \frac{W_t}{A_t P_t} \Rightarrow \hat{m}c_t = \hat{\omega}_t - a_t$$

- Labor market clearing:

$$N_t = \frac{Y_t}{A_t} D_t \Rightarrow \hat{n}_t = \hat{y}_t - a_t$$

- Again, here we have that $\hat{a}_t = a_t$

Full system with TFP shocks

- The log-linearized equilibrium is characterized by

$$\text{Intratemporal hh optimality: } \hat{\omega}_t = \hat{c}_t + \varphi \hat{n}_t$$

$$\text{Intertemporal hh optimality: } \hat{c}_t = -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{c}_{t+1}$$

$$\text{Firm optimality: } \pi_t = \beta E_t \pi_{t+1} + \lambda \widehat{mc}_t$$

$$\text{Marginal cost: } \widehat{mc}_t = \hat{\omega}_t - a_t$$

$$\text{Goods clearing: } \hat{c}_t = \hat{y}_t$$

$$\text{Labor clearing: } \hat{y}_t = \hat{n}_t + a_t$$

$$\text{Policy: } \hat{i}_t = \phi \pi_t + \nu_t$$

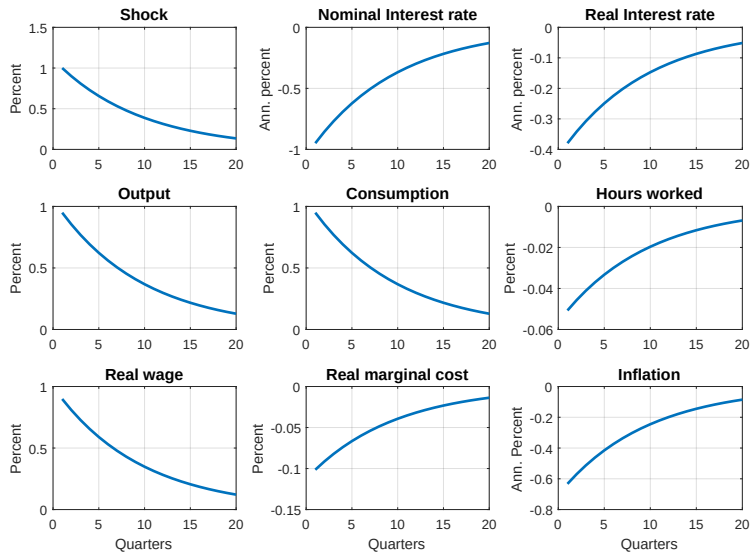
where $\hat{\omega}_t = \hat{w}_t - p_t$ denotes log deviations in the real wage

- Also, the law of motion for exogenous shocks:

$$\nu_t = \rho_\nu \nu_{t-1} + \epsilon_t$$

$$a_t = \rho_a a_{t-1} + \epsilon_t^a$$

IRFs to a TFP shock



Mechanism

- Again, no hump shapes like in the vanilla RBC model.
 - ▶ Again, without capital, there is no state variable.
- Recall, in RBC without capital, hours worked did not respond at all
- Here, hours worked falls
 - ▶ Galí (AER 1999) and Basu-Fernald-Kimball (AER 2006) argue this is consistent with the data
- What role do sticky prices and monetary policy play for this response?
- Again, let's go back to flexible prices (which is just the RBC model with no capital (and monopolistic competition))
- With flexible prices, firm optimality is

$$\begin{aligned} P_{it} &= M\psi_t \\ \frac{P_{it}}{P_t} &= M\frac{\psi_t}{P_t} \Rightarrow \widehat{mc}_t = 0 \end{aligned}$$

- All other relationships are unaffected

Equilibrium system with flexible prices

- The log-linearized equilibrium is characterized by

Intratemporal hh optimality: $\hat{\omega}_t = \hat{c}_t + \varphi \hat{n}_t$

Intertemporal hh optimality: $\hat{c}_t = -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{c}_{t+1}$

Firm optimality: $\widehat{mc}_t = 0$

Marginal cost: $\widehat{mc}_t = \hat{\omega}_t - a_t$

Goods clearing: $\hat{c}_t = \hat{y}_t$

Labor clearing: $\hat{y}_t = \hat{n}_t + a_t$

Policy: $\hat{i}_t = \phi \pi_t + \nu_t$

where $\hat{\omega}_t = \hat{w}_t - p_t$ denotes log deviations in the real wage

- Also, the law of motion for exogenous shocks:

$$\nu_t = \rho_\nu \nu_{t-1} + \epsilon_t$$

$$a_t = \rho_a a_{t-1} + \epsilon_t^a$$

Equilibrium system with flexible prices: solution

- The log-linearized equilibrium is characterized by

$$\text{Intratemporal hh optimality: } \hat{\omega}_t = \hat{c}_t + \varphi \hat{n}_t$$

$$\text{Intertemporal hh optimality: } \hat{c}_t = -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{c}_{t+1}$$

$$\text{Firm optimality: } \widehat{mc}_t = 0$$

$$\text{Marginal cost: } \widehat{mc}_t = \hat{\omega}_t - a_t$$

$$\text{Goods clearing: } \hat{c}_t = \hat{y}_t$$

$$\text{Labor clearing: } \hat{y}_t = \hat{n}_t + a_t$$

$$\text{Policy: } \hat{i}_t = \phi \pi_t + \nu_t$$

- Work it through to find the natural interest rate: (Do on whiteboard)

$$\begin{aligned} \hat{r}_t^n &= \hat{i}_t - E_t \pi_{t+1} \\ &= -(1 - \rho_a) a_t \end{aligned}$$

- Key result: positive TFP shocks cause a decline in the natural real interest rate
- Mechanism: $a_t > 0 \Rightarrow \Delta E_t a_{t+1} < 0 \Rightarrow \Delta E_t c_{t+1} < 0 \Rightarrow \hat{r}_t^n < 0$

Sticky-price equilibrium in gaps

- Let's turn back to the **sticky-price equilibrium**
- We linearized the equilibrium around the steady state $\hat{x} = x_t - x_{ss}$, for any x
- Let's instead consider a linearization around the **flexible-price equilibrium** or, using another terminology, the **natural equilibrium**
- We'll go directly at the 3-equation system
- Define “gaps” as $\tilde{x}_t = x_t - x_t^n$
- Note $\tilde{x}_t = (x_t - x_{ss}) - (x_t^n - x_{ss}) = \hat{x}_t - \hat{x}_t^n$

Sticky-price equilibrium in gaps II

- DIS curve: not affected by a_t , simply subtract natural equilibrium to find

$$\begin{aligned}\hat{y}_t &= -(\hat{i}_t - E_t \pi_{t+1}) + E_t \hat{y}_{t+1} \\ \Leftrightarrow \hat{y}_t - \hat{y}_t^n &= -(\hat{i}_t - E_t \pi_{t+1} - \hat{r}_t^n) + E_t \hat{y}_{t+1} - E_t \hat{y}_{t+1}^n \\ \Leftrightarrow \tilde{y}_t &= -(\hat{i}_t - E_t \pi_{t+1} - \hat{r}_t^n) + E_t \tilde{y}_{t+1}\end{aligned}$$

- Phillips curve:

- ▶ Start with firm optimality:

$$\pi_t = \beta E_t \pi_{t+1} + \lambda \widehat{mc}_t$$

- ▶ Flexible prices: $\widehat{mc}_t^n = 0 \Rightarrow \widehat{mc}_t = \widetilde{mc}_t$
- ▶ Work through the flex-price equations to find $\widetilde{mc}_t = (1 + \varphi)\tilde{y}_t$, hence

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t$$

where $\kappa = \lambda(1 + \varphi)$

Sticky-price equilibrium in gaps III

- Putting it together, the equilibrium can be summarized as

$$\text{DIS curve:} \quad \tilde{y}_t = -(\hat{i}_t - E_t \pi_{t+1} - \hat{r}_t^n) + E_t \tilde{y}_{t+1}$$

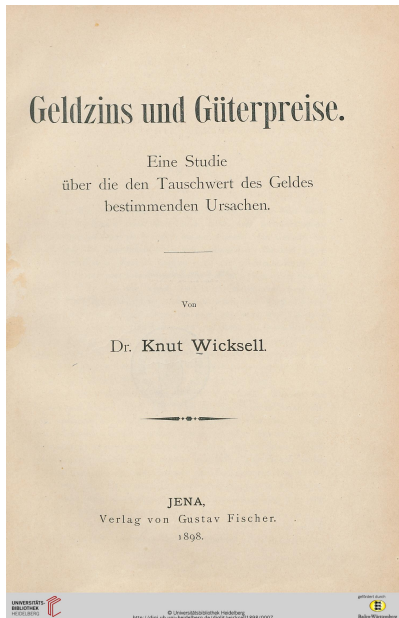
$$\text{Phillips curve:} \quad \pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t$$

$$\text{Policy rule:} \quad \hat{i}_t = \phi \pi_t + \nu_t$$

$$\text{Natural real interest rate:} \quad \hat{r}_t^n = -(\rho_a - 1)a_t$$

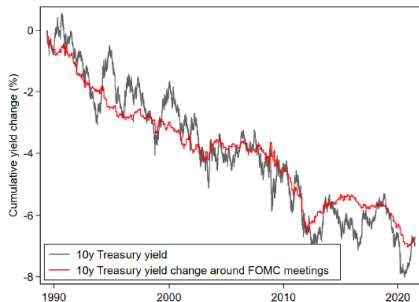
- $\tilde{y}_t$ measures fluctuations that are **inefficient**
- Model captures two central assertions of the prevailing **Neo-Wicksellian** view of business cycles
 - Monetary factors do not affect the natural real interest rate
 - Inefficient fluctuations are linked to deviations of the real interest rate from the natural real interest rate

Intermezzo: The Great Knut Wicksell

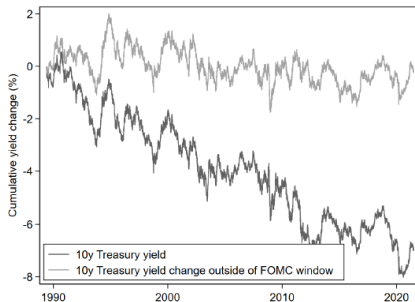


Intermezzo: Is the natural-rate hypothesis correct?

(A) 3-day window around FOMC meetings

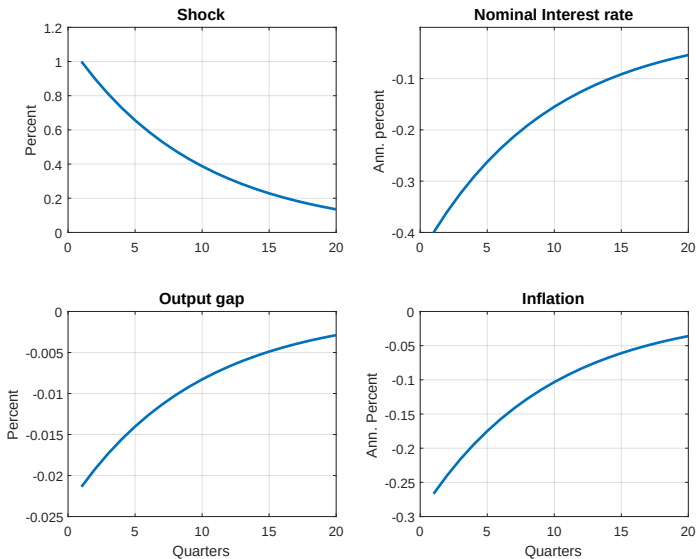


(B) Days outside 3-day FOMC window



Source: Hillenbrand (RFS 2024)

IRFs to a TFP shock: Gaps



IRFs to a TFP shock: comments

- A positive TFP shock leads to a “gap” recession
- This reflects that monetary policy is not doing a good job
- Mechanism: $TFP \uparrow \Rightarrow \text{marginal cost} \downarrow \Rightarrow \text{inflation} \downarrow \Rightarrow \text{interest rate} \downarrow$
 - ▶ Monetary policy “stimulates” the economy...
 - ▶ ... but not enough to raise consumption to its efficient level
 - ▶ In the “natural equilibrium”, real interest rate is even lower, and hours worked are constant
 - ▶ Therefore, hours decline in the observed equilibrium
- This inefficiency is not surprising: the policy rule was specified in a completely ad hoc manner

Cost-push shocks

Cost-push shocks

- TFP shocks are an example of a *supply shock*
- Due to sticky prices, TFP shocks create a wedge (“deviations”) between the **natural equilibrium** and the actual equilibrium
- Because fluctuations in the natural equilibrium are efficient, the policy problem tends to be organized around how to undo the effect of price stickiness
- Another class of supply shocks creates a wedge between the **natural equilibrium** and the **efficient equilibrium**, the so-called **cost-push shocks**
- Examples: shocks to firms’ desired markups (greedflation?), shocks to distortionary taxes

Rewriting the system once more

- Written in deviations from the natural equilibrium, our system is

$$\begin{aligned}\text{DIS curve:} \quad & \tilde{y}_t = -(\hat{l}_t - E_t \pi_{t+1} - \hat{r}_t^n) + E_t \tilde{y}_{t+1} \\ \text{Phillips curve:} \quad & \pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t \\ \text{Policy rule:} \quad & \hat{l}_t = \phi \pi_t + \nu_t\end{aligned}$$

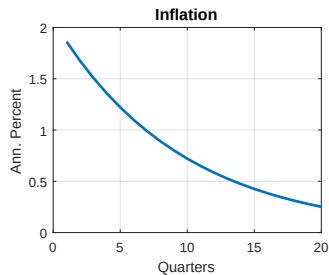
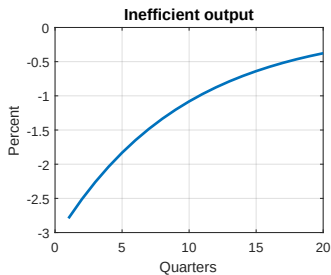
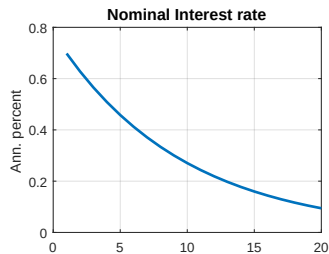
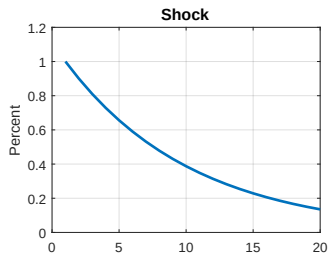
- Define $x_t = y_t - y_t^e$, where y_t^e is the efficient equilibrium path.
- We have that $\tilde{y}_t = x_t + (y_t^e - y_t^n)$
- We can write our system as

$$\begin{aligned}\text{DIS curve:} \quad & x_t = -(\hat{l}_t - E_t \pi_{t+1} - \hat{r}_t^e) + E_t x_{t+1} \\ \text{Phillips curve:} \quad & \pi_t = \beta E_t \pi_{t+1} + \kappa x_t + u_t \\ \text{Policy rule:} \quad & \hat{l}_t = \phi \pi_t + \nu_t\end{aligned}$$

where $u_t = \kappa(y_t^e - y_t^n)$ and $\hat{r}_t^e = E_t \Delta y_{t+1}^e$

- u_t is, in *reduced form*, a cost-push shock

IRFs to a cost-push shock



- Cost-push shocks are interesting from a policy perspective
- These shocks increase inflation, but also cause an inefficient recession
 - ▶ A monetary authority can combat inflation, but this will lead an even deeper recession
- Contrast with a positive TFP shock: inefficient recession, but lowers inflation
 - ▶ Here, a monetary authority can seemingly address two problems simultaneously by stimulating the economy
- It seems like we need to think more systematically about the policy problem...

Summing up

- Basic NK model = RBC model + monopolistic competition and sticky prices
- In contrast to RBC, NK predicts inefficient fluctuations and a role for monetary policy
- In response to a positive shock, the NK model predicts that

Shock	$\hat{y}_t$	$\tilde{y}_t$	x_t	π_t
Monetary policy	down	down	down	down
TFP	up	down	down	down
Cost-push	—	—	down	up

- In line with the evidence on monetary policy (qualitatively, but not quantitatively)
- Next lecture: optimal policy in the NK model